

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 10: Arithmetic Progression

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Sum of n terms

1. An A.P. has first term 5 and last term 50, and its terms add up to 440. Find (a) the number of terms (b) the common difference. **ICSE 2025, 3 marks**

SOLUTION

$$(a) S = \frac{n}{2}(a + l): 440 = \frac{n}{2}(5 + 50), \text{ so } n = \frac{880}{55} = 16.$$

$$(b) l = a + (n - 1)d: 50 = 5 + 15d, \text{ so } d = 3.$$

Answer: (a) 16 terms (b) common difference 3

2. The n th term of an A.P. is $T_n = 6(7 - n)$. Find (a) its first term and common difference (b) the sum of its first 25 terms. **ICSE 2024, 3 marks**

SOLUTION

$$(a) T_1 = 6 \times 6 = 36 \text{ and } T_2 = 6 \times 5 = 30, \text{ so } a = 36 \text{ and } d = -6.$$

$$(b) S_{25} = \frac{25}{2} [2(36) + 24(-6)] = \frac{25}{2} \times (-72)$$

Answer: (a) $a = 36, d = -6$ (b) $S_{25} = -900$

3. The 5th term of an A.P. is 4 and its 9th term is -12 . Find (a) the first term (b) the common difference (c) the sum of the first 16 terms. **ICSE 2023, 3 marks**

SOLUTION

$$a + 4d = 4 \text{ and } a + 8d = -12. \text{ Subtracting: } 4d = -16, \text{ so } d = -4 \text{ and } a = 20.$$

$$S_{16} = \frac{16}{2} [2(20) + 15(-4)] = 8 \times (-20)$$

Answer: (a) 20 (b) -4 (c) -160

4. Which term of the A.P. 15, 30, 45, 60, ... is 300? Find the sum of all the terms up to it. **ICSE 2023, 3 marks**

SOLUTION

$$a = 15, d = 15. 15 + (n - 1) \times 15 = 300, \text{ so } 15n = 300 \text{ and } n = 20.$$

$$S_{20} = \frac{20}{2}(15 + 300) = 10 \times 315$$

Answer: the 20th term; sum = 3150

- 5.** The 6th term of an A.P. is four times its first term, and the sum of its first six terms is 75. Find the first term and the common difference. **ICSE 2020, 3 marks**

SOLUTION

$$a + 5d = 4a, \text{ so } 5d = 3a.$$

$$S_6 = \frac{6}{2}(2a + 5d) = 3(2a + 3a) = 15a = 75, \text{ so } a = 5.$$

$$5d = 15, \text{ so } d = 3.$$

Answer: first term 5, common difference 3

- 6.** The 4th and 6th terms of an A.P. are 8 and 14. Find (a) the first term (b) the common difference (c) the sum of the first 20 terms. **ICSE 2019, 4 marks**

SOLUTION

$$a + 3d = 8 \text{ and } a + 5d = 14. \text{ Subtracting: } 2d = 6, \text{ so } d = 3 \text{ and } a = -1.$$

$$S_{20} = \frac{20}{2} [2(-1) + 19(3)] = 10 \times 55$$

Answer: (a) -1 (b) 3 (c) 550

- 7.** The 4th term of an A.P. is 22 and the 15th term is 66. Find the first term and the common difference, and then the sum of the first 8 terms. **ICSE 2018, 4 marks**

SOLUTION

$$a + 3d = 22 \text{ and } a + 14d = 66. \text{ Subtracting: } 11d = 44, \text{ so } d = 4 \text{ and } a = 10.$$

$$S_8 = \frac{8}{2} [2(10) + 7(4)] = 4 \times 48$$

Answer: $a = 10, d = 4, S_8 = 192$

2 The general term

- 8.** Find the 7th term of the A.P. $\frac{1}{a}, \frac{1}{a} + 1, \frac{1}{a} + 2, \dots$ (a) $\frac{1}{a} + 6$ (b) $\frac{1}{a} + 7$ (c) $\frac{1}{a} + 8$ (d) $\frac{1}{a} + 7^7$

ICSE 2024, 1 mark

SOLUTION

$$\text{First term} = \frac{1}{a} \text{ and } d = 1. T_7 = \frac{1}{a} + 6 \times 1$$

Answer: (a)

- 9.** The n th term of an A.P. is $2n + 5$. Find its 10th term. (a) 7 (b) 15 (c) 25 (d) 45 **ICSE 2023, 1 mark**

SOLUTION

$$T_{10} = 2(10) + 5 = 25$$

Answer: (c)

- 10.** $(k - 3)$, $(2k + 1)$ and $(4k + 3)$ are three consecutive terms of an A.P. Find k . **ICSE 2018, 3 marks**

SOLUTION

The middle term is the average of the other two: $2(2k + 1) = (k - 3) + (4k + 3)$

$$4k + 2 = 5k$$

Answer: $k = 2$

3 Choosing terms conveniently

- 11.** The first three terms of an A.P. add up to 42, and the product of the first and third terms is 52. Find the first term and the common difference. **ICSE 2019, 3 marks**

SOLUTION

Let the terms be $a - d$, a , $a + d$. Their sum is $3a = 42$, so $a = 14$.

$$(14 - d)(14 + d) = 52, \text{ so } d^2 = 144 \text{ and } d = \pm 12.$$

Answer: first term 2 with common difference 12, or first term 26 with common difference -12

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